

Efficiency

relative, impact per unit of work

Sufficiency

absolute, total burden



Step 1: Wider ranking

Step 2: Sufficiency indicators

Step 3: What-if scenarios

Rank systems on carbon, water, and embodied impact beyond watts alone

Account each system's absolute burden: lifecycle total, utilization, replacement cycle

Re-rank under change: cleaner grid, newer hardware, drier cooling, rising demand

Publicly reported

Top500 and Green500 ranks

Peak performance

Benchmark-time power

Published impact factors

Not reported

Actual power draw

Utilization

Embodied impacts